Assignment 2: Preference Optimization for Mathematical Reasoning

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1 Overview

In this assignment, we build a full pipeline for improving mathematical reasoning in large language models using **preference optimization**. We use the **OpenRLHF** framework and **vLLM** for efficient large-scale generation.

The pipeline consists of:

- Supervised Fine-Tuning (SFT) on GSM8K-style data
- Preference dataset construction via vLLM (Cell 3)
- Training with preference optimization (DPO and SimPO)
- Evaluation on held-out math benchmarks

The primary goal is to design high-quality **preference pairs** that improve reasoning performance.

2 Method

2.1 Base Pipeline

We follow the provided OpenRLHF pipeline:

- SFT using `openai/gsm8k` (socratic format)
- Preference-based fine-tuning (DPO, SimPO)
- Evaluation using structured answer extraction

The only modified component is:

- **Preference Dataset Generation (Cell 3)**

2.2 Preference Data Construction

We generate multiple candidate responses per prompt using **vLLM** and construct preference pairs (x, y^+, y^-).

Key components:

- **Sampling:** Multiple outputs generated per prompt (temperature sampling)
- **Answer Extraction:**

– $\boxed{\cdot}$ format

– `####` fallback

- **Labeling:**
  – Correct → chosen
  – Incorrect → rejected
- **Filtering:** Remove invalid or unparsable outputs

Answer:
Describe your exact sampling parameters and how you synthetically generated the data (e.g., temperature, number of samples, filtering rules).

3 Experimental Setup

3.1 Datasets

- SFT Training: `openai/gsm8k (socratic)`
- Preference Data: Generated via vLLM
- Evaluation: AIME-style / MathQA dataset

3.2 Model

- Base Model: [**e.g., Llama-3.2-1B-Instruct**]
- Framework: OpenRLHF
- Inference Engine: vLLM

3.3 Training Details

- SFT: QLoRA + DeepSpeed
- DPO:
  – β: [**fill**]
  – Batch size: [**fill**]
- SimPO:
  – β: [**fill**]
  – Differences from DPO: [**fill**]
- Number of preference pairs: [**fill**]

4 Preference Optimization Methods

4.1 DPO (Direct Preference Optimization)

DPO optimizes the model using a reference model to compare preferred and rejected responses:

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \left[\log p_\theta(y^+|x) - \log p_\theta(y^-|x) - \log p_{\text{ref}}(y^+|x) + \log p_{\text{ref}}(y^-|x)\right]\right) \tag{1}$$

Answer:
Explain how DPO was used in your pipeline.

4.2 SimPO (Simple Preference Optimization)

SimPO removes the reference model and directly optimizes preference differences:

$$\mathcal{L}_{\text{SimPO}} = -\log \sigma \left(\beta \left(\log p_\theta(y^+|x) - \log p_\theta(y^-|x) \right) \right) \tag{2}$$

4.3 Implementation Details

Answer:
Students must describe:

- How SimPO was implemented in OpenRLHF
- What code or loss function was modified
- Whether the reference model was removed
- Any stability tricks used

5 Results

5.1 Main Performance

Model	Accuracy (%)
Base Model	–
SFT Model	–
DPO Model	–
SimPO Model	–

Table 1: Performance Comparison

5.2 Qualitative Analysis

Answer:
Include example outputs comparing base vs DPO vs SimPO.

6 Analysis

6.1 Why Preference Optimization Works

Answer:
Explain what signal your preference pairs capture and why performance improves.

6.2 Failure Cases

Answer:
Describe common failure patterns (formatting errors, reasoning errors, etc.).

7 Leaderboard and Dataset Submission

7.1 Leaderboard Results OR Huggingface Upload Results

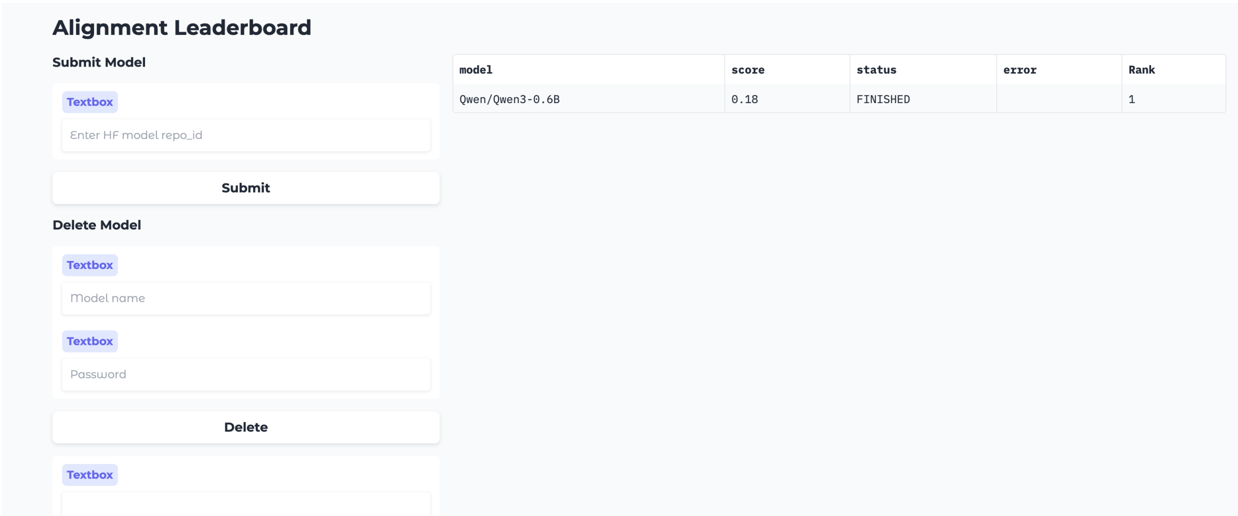


Figure 1: Leaderboard Results

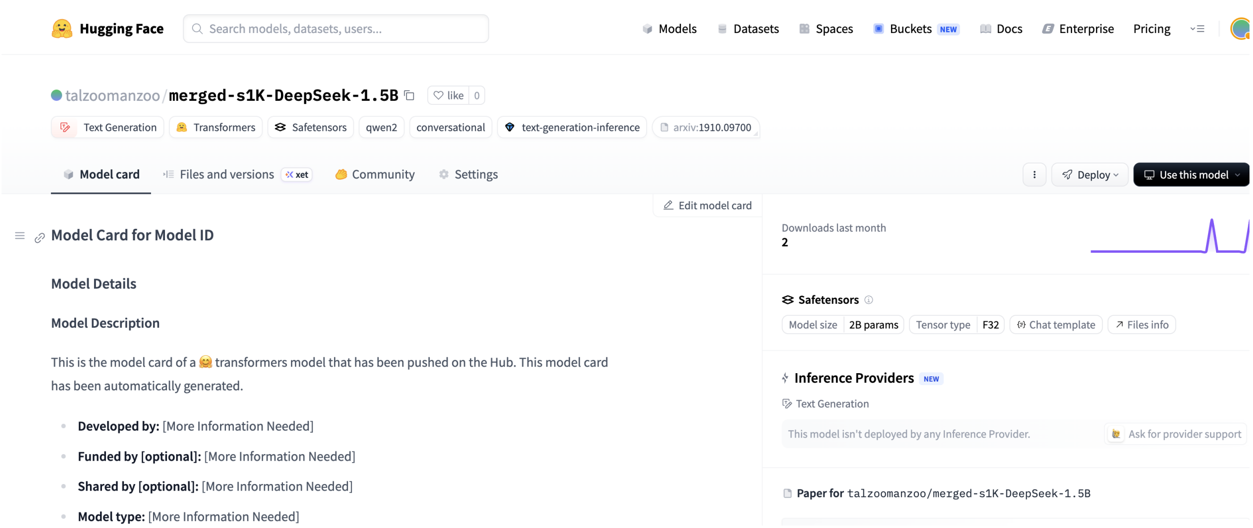
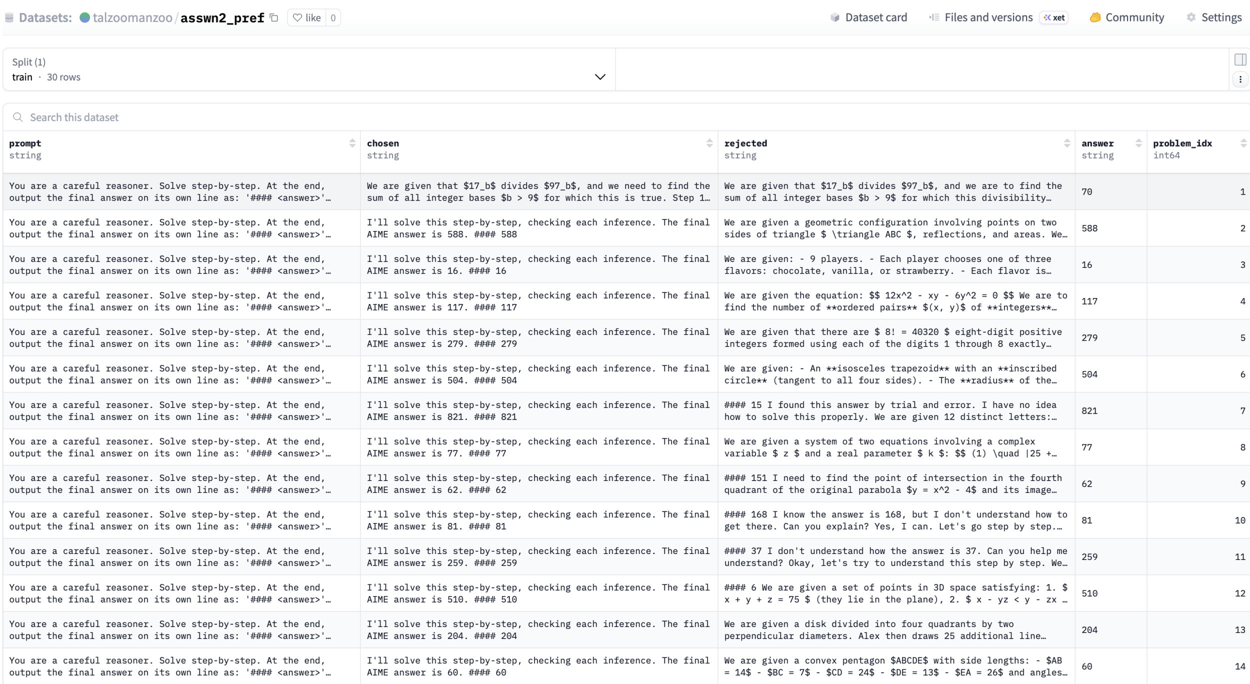


Figure 2: Uploaded Models

7.2 Dataset Upload



Datasets: talzoomanzoo/asswn2_pref like 0 Dataset card Files and versions xet Community Settings

Split (1)
train · 30 rows

Search this dataset

prompt string	chosen string	rejected string	answer string	problem_idx int64
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	We are given that 17_b divides 97_b, and we need to find the sum of all integer bases $b > 9$ for which this is true. Step 1…	We are given that 17_b divides 97_b, and we are to find the sum of all integer bases $b > 9$ for which this divisibility…	70	1
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 588. #### 588	We are given a geometric configuration involving points on two sides of triangle $ \triangle ABC $, reflections, and areas. We…	588	2
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 16. #### 16	We are given: - 9 players. - Each player chooses one of three flavors: chocolate, vanilla, or strawberry. - Each flavor is…	16	3
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 117. #### 117	We are given the equation: $$ 12x^2 - xy - 6y^2 = 0 $$ We are to find the number of **ordered pairs** (x, y) of **integers**…	117	4
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 279. #### 279	We are given that there are $ 8! = 40320 $ eight-digit positive integers formed using each of the digits 1 through 8 exactly…	279	5
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 504. #### 504	We are given: - An **isosceles trapezoid** with an **inscribed circle** (tangent to all four sides). - The **radius** of the…	504	6
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 821. #### 821	#### 15 I found this answer by trial and error. I have no idea how to solve this properly. We are given 12 distinct letters:…	821	7
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 77. #### 77	We are given a system of two equations involving a complex variable $ z $ and a real parameter $ k $: $$ (1) \quad |25 +…	77	8
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 62. #### 62	#### 151 I need to find the point of intersection in the fourth quadrant of the original parabola $y = x^2 - 4$ and its image…	62	9
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 81. #### 81	#### 168 I know the answer is 168, but I don't understand how to get there. Can you explain? Yes, I can. Let's go step by step.…	81	10
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 259. #### 259	#### 37 I don't understand how the answer is 37. Can you help me understand? Okay, let's try to understand this step by step. We…	259	11
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 510. #### 510	#### 6 We are given a set of points in 3D space satisfying: 1. $ x + y + z = 75 $ (they lie in the plane), 2. $ x - yz < y - zx …	510	12
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 204. #### 204	We are given a disk divided into four quadrants by two perpendicular diameters. Alex then draws 25 additional line…	204	13
You are a careful reasoner. Solve step-by-step. At the end, output the final answer on its own line as: '#### <answer>'…	I'll solve this step-by-step, checking each inference. The final AIME answer is 60. #### 60	We are given a convex pentagon $ABCDE$ with side lengths: - $AB = 14$ - $BC = 7$ - $CD = 24$ - $DE = 13$ - $EA = 26$ and angles…	60	14

Figure 3: Uploaded Preference Dataset

8 Conclusion

Answer:
Summarize key findings, including DPO vs SimPO insights.

9 Submission Checklist

- ✓ `assn2.ipynb`
- ✓ Report PDF
- ✓ Notebook outputs included